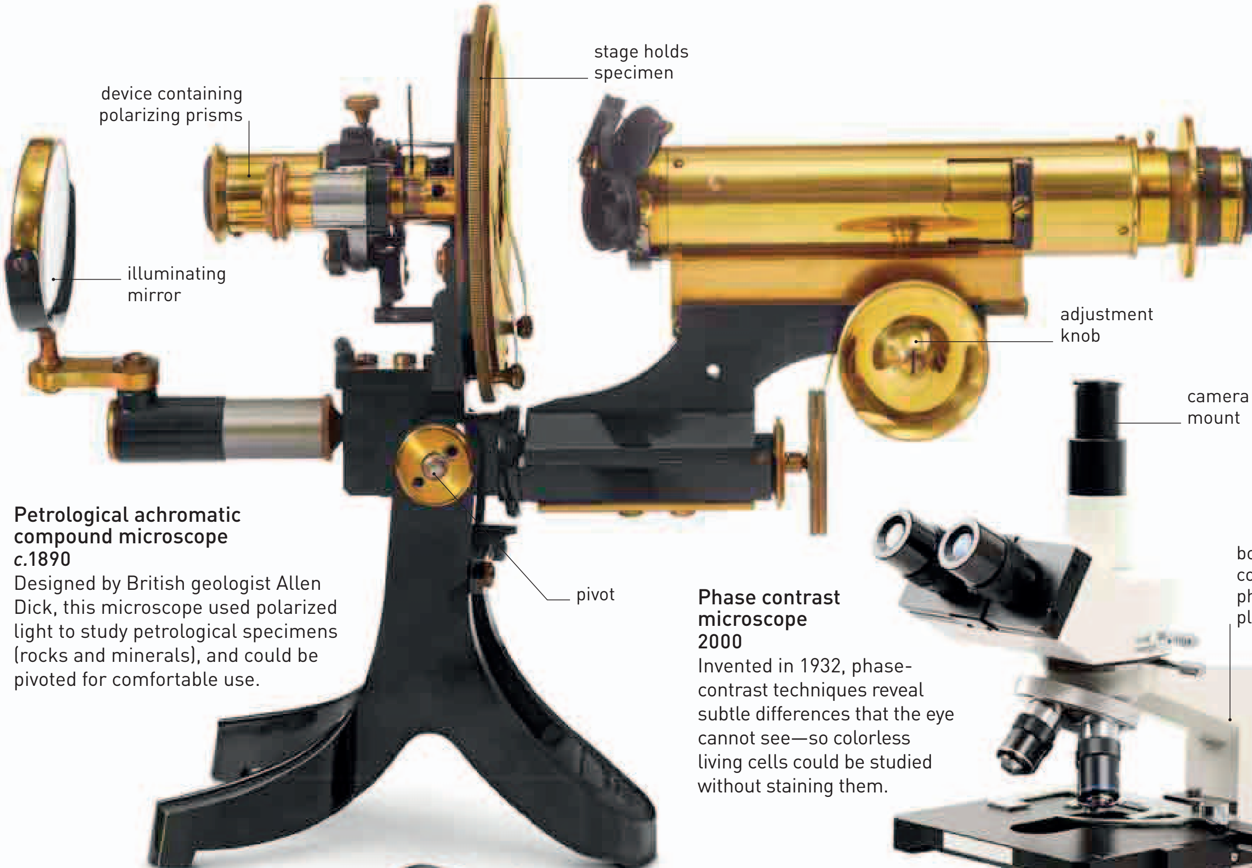
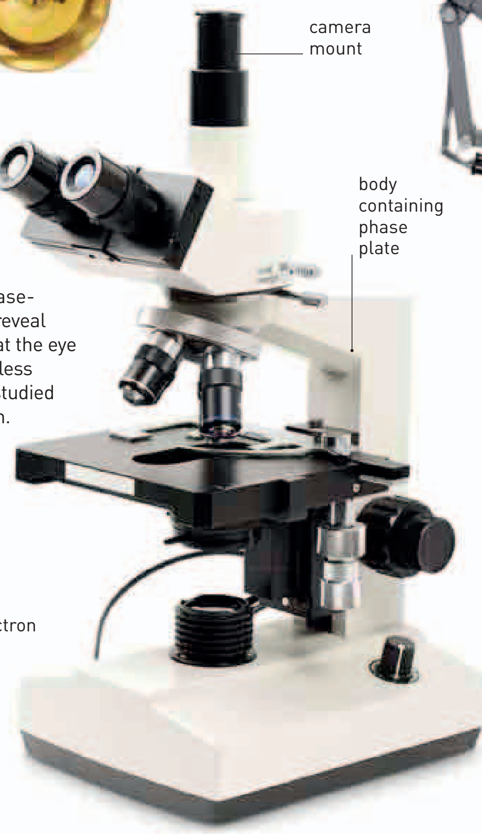


Petrological achromatic compound microscope
*c.*1890
Designed by British geologist Allen Dick, this microscope used polarized light to study petrological specimens (rocks and minerals), and could be pivoted for comfortable use.

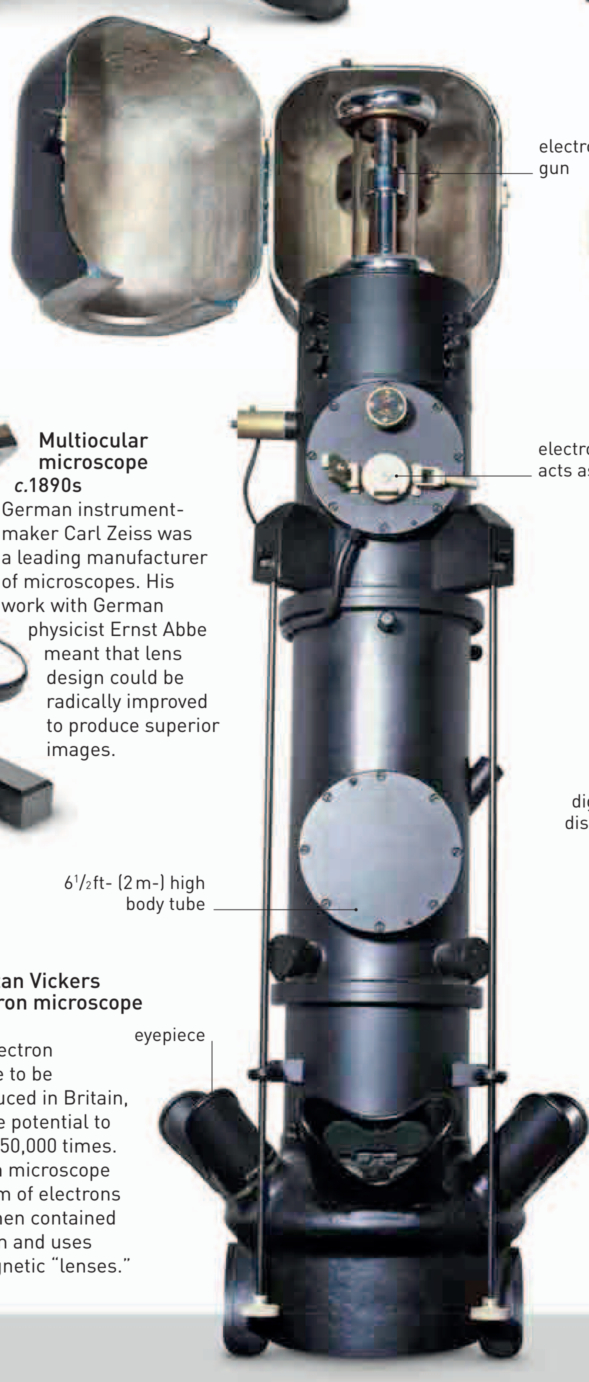
Polarizing light microscope
*c.*1980
Polarizing filters—which line up light vibrations in one direction—are used in microscopes to study the optical properties of crystal.



Phase contrast microscope
2000
Invented in 1932, phase-contrast techniques reveal subtle differences that the eye cannot see—so colorless living cells could be studied without staining them.

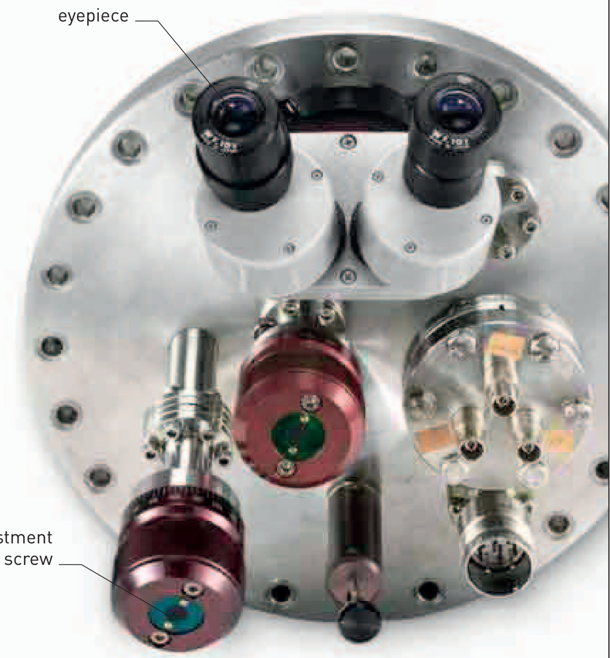
USB microscope
2008
A Universal Serial Bus (USB) microscope is a miniature device that is connected to a computer to generate on-screen images of magnified specimens.

Multiocular microscope
*c.*1890s
German instrument-maker Carl Zeiss was a leading manufacturer of microscopes. His work with German physicist Ernst Abbe meant that lens design could be radically improved to produce superior images.



Metropolitan Vickers EM2 electron microscope
*c.*1946
The first electron microscope to be mass produced in Britain, this had the potential to magnify to 50,000 times. An electron microscope fires a beam of electrons at a specimen contained in a vacuum and uses electromagnetic "lenses."

Atomic force microscope
*c.*2000
Developed from the scanning tunneling microscope in 1986, this scans objects with an atom-sized probe and is one of the most powerful microscopes available today.



Scanning tunneling microscope
1986
Invented in 1981, this was the first kind of microscope that allowed scientists to see individual atoms. Objects could be viewed to a resolution of a nanometer (one millionth of a millimeter).